Mach-Zehnder Interferometer for SiEPIC Course

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***Abstract*—**

***Index Terms*—**

I. Introduction

THE Mach-Zehnder Interferometer (MZI) is a passive device that has many applications, namely modulation for high speed optical communications and optical filters.

II. Theory

In the most basic form, the input of the MZI is split via a Y-branch, travels through two separate waveguides, and then recombined via another Y-branch at the output. Other variants exist that utilize other types of splitters/combiners such as adiabatic splitters, bidirectional couplers, or custom splitters. The proceeding theory analyzes a basic MZI.

[insert Y-branch diagram with port labelling 1, 2, and 3]

When light is propagated through input port 1, the power intensity of the input, Ii, is split in half in output ports 2 and 3, $I_2 = I_3 = \frac{I_1}{2}$. Since $I \propto |E|^2$, the E fields in output ports 2 and 3 resolve to

$$E_o = \sqrt{I_o} = \sqrt{\frac{I_i}{2}} = \frac{E_i}{\sqrt{2}}$$

Additionally, the propagation constant in a waveguide is described by $\beta = \frac{2\pi n_{eff}}{\lambda}$. The propagation constant describes how a wave's phase changes through a material along a distance. It is a function of wavelength, as is the index of refraction n, and is derived from Maxwell's equations. To describe the loss of power intensity per unit length through a waveguide, α is used. However, the loss coefficient is halved for the E field since

$$I = Ie^{-i\beta z + \alpha z} \propto |E|^2 = |Ee^{-i\beta z + \frac{\alpha}{2} z}|^2$$

. This means that for two waveguides after the first Y-branch of lengths L_1 and L_2, the two E fields in the branches are

$$E_1 = \frac{E_i}{\sqrt{2}} e^{-(i\beta_1 L_1 + \frac{\alpha_1}{2} L_1)}$$

$$E_2 = \frac{E_i}{\sqrt{2}} e^{-(i\beta_2 L_2 + \frac{\alpha_2}{2} L_2)}$$

The two waveguides are then combined at an output Y-branch. Using the relationship between E_in and E_out derived above, E_o at the combiner is

$$E_o = \frac{E_i}{\sqrt{2}} = \frac{E_1 + E_2}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \frac{E_i}{\sqrt{2}} (e^{-(i\beta_1 L_1 + \frac{\alpha_1}{2} L_1)} + e^{-(i\beta_2 L_2 + \frac{\alpha_2}{2} L_2)})$$

Converting to power intensity yields

$$I_o = |E_o|^2 = \frac{I_i}{4} |e^{-(i\beta_1 L_1 + \frac{\alpha_1}{2} L_1)} + e^{-(i\beta_2 L_2 + \frac{\alpha_2}{2} L_2)}|^2$$

Squaring the inside of the |.| with it's complex conjugate and assuming losses are negligible ($\alpha = 0$) yields

$$2 + 2\cos(\beta_1 L_1 - \beta_2 L_2)$$

Therefore, I_o simplifies to

$$I_o = \frac{I_i}{2}[1 + \cos(\beta_1 L_1 - \beta_2 L_2)]$$

The transfer function, or transmission, of an MZI is defined as the ratio of the output over the input

$$T = \frac{I_o}{I_i} = \frac{1}{2}[1 + \cos(\beta_1 L_1 - \beta_2 L_2)]$$

This transfer function produces Lorentzian dips that are periodic over frequency (and quasi-periodic over wavelength), see Fig. XX as an example. The differnce in wavlength between two neighboring dips is called the full spectral range (FSR), and this is typically a metric of much interest when designing MZIs, especially for modulation and filtering.

$$FSR = \Delta\lambda = \lambda_{m+1} - \lambda_m$$

Given the transfer function of the MZI, the FSR can be derived. First, it is assumed that β in each branch of the MZI is equal (i.e. n_{eff} and λ are the same in both branches). Ideally, a dip occurs every time the transmission is 0 (the dips are essentially notch filters), so

$$0 = \frac{1}{2}[1 + \cos(\beta \Delta L)$$

$$\beta \Delta L = \cos^{-1}(-1) = \pi$$

Since this happens at every 2π spacing,

$$\Delta\beta = \frac{2\pi}{\Delta L} \approx -\Delta\lambda \frac{d\beta}{d\lambda}$$

Despite $\beta \propto \frac{n_{eff}(\lambda)}{\lambda}$, $\Delta\beta$ can be linearly approximated over wavelength assuming a small enough FSR when compared to the ΔL, which is true for this design run. Solving for $\Delta\lambda$ above,

$$\Delta\lambda = -\frac{2\pi}{\Delta L} (\frac{d\beta}{d\lambda})^{-1}$$

$$\frac{d\beta}{d\lambda} = \frac{d}{d\lambda} (\frac{2\pi n_{eff}(\lambda)}{\lambda})$$

$$= 2\pi \frac{\lambda \frac{dn}{d\lambda} - n}{\lambda^2}$$

Finally, applying this to the $\Delta\lambda$ equation

$$FSR = \frac{\lambda^2}{(n - \lambda \frac{dn}{d\lambda})\Delta L} = \frac{\lambda^2}{n_g \Delta L}$$

Where $n_g = n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}$

III. Modeling and Simulation

Lumerical MODE was used to simulate the waveguides to examine the quasi-TE and quasi-TM fundamental modes in a 500 nm x 220 nm x 5 um (x, y, z) where z is the direction of propagation. The compact waveguide equation for a wavlength of 1550 nm is

$$n_{eff}(\lambda) = 2.45 - 1.13(\lambda - 1,55) - 0.044(\lambda - 1.55)^2$$

A. Quasi-TE Mode WG

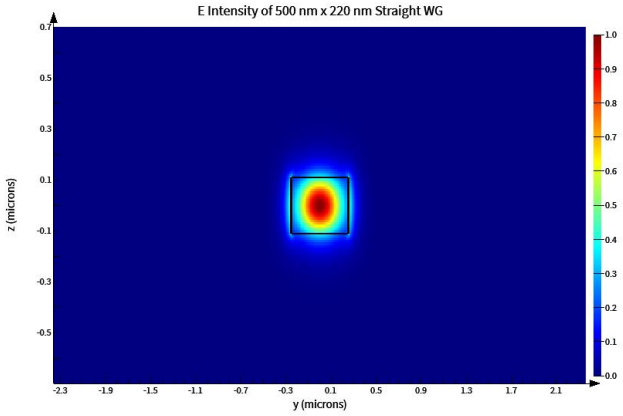


Fig. 1. E intensity of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TE mode 1.

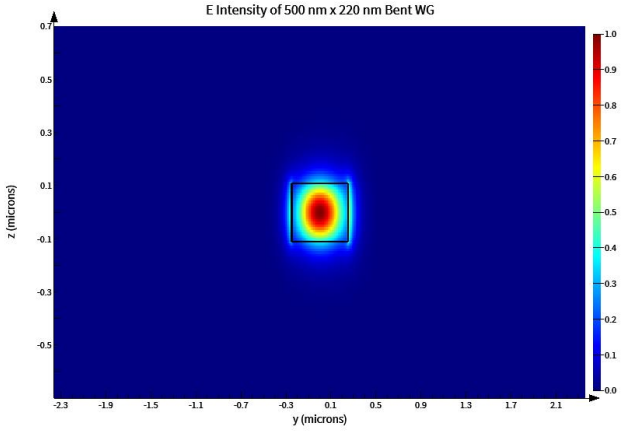


Fig. 2. E intensity of a 500 nm x 220 nm crossection of a bent Si waveguide in SiO2 cladding of the quasi-TE mode 1.

Understanding the modes propagating through the waveguide verifies that the choosen waveguide can perform optimally for the desired mode (in this case, TE), and that the width is not too small that the power coupling is too small, but not so large that it introduces large power coupling of additional modes (i.e. TE1, TE2, TE3, etc.). An overlap analysis can also be performed to analyze the impacts of the bend radius on losses. For this simulation, a 5 μm x 500 nm x 220 nm (x, y, z) straight waveguide was used with a FDE z-span of 1.4 μm, since the convergence test on mode-mismatch loss shows that the FDE boundary converges around 1.4 μm for TE mode. When using the default bend radius of 5 μm, the power coupling into the bent waveguide is 0.998625, or approximately 0.14%, or XX dB of loss. The offset of the two waveguides to the bend can be optimized to achieve the minimum loss. [insert pics of offset analysis] The optimization algorithm found that an offset of 10 nm can improve the loss to 0.999483, which is 0.06% or XX dB of loss. Therefore, in the layout, some MZI designs have no offset while others do.

Lumerical MODE was used to calculate n_{eff} and n_g vs. wavelength.

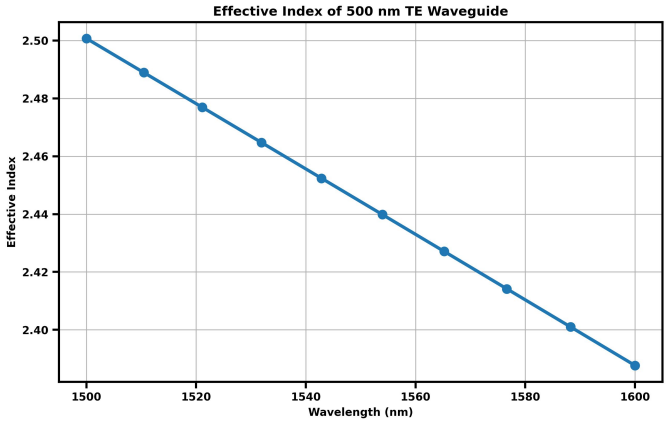


Fig. 3. n_{eff} vs wavelength of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TE mode 1.

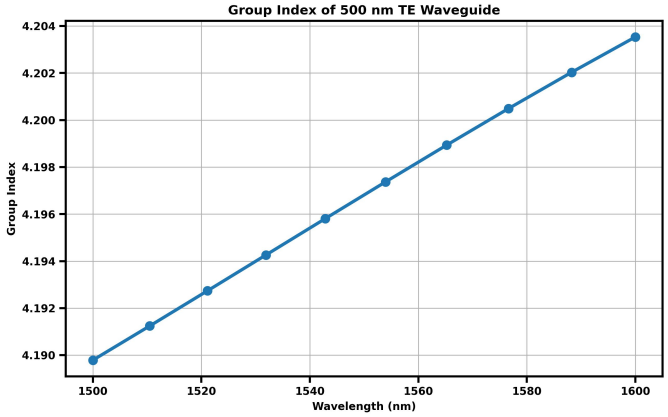


Fig. 4. n_g of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TE mode 1.

B. Quasi-TM Mode WG

The quasi-TM mode was also analyzed using Lumerical MODE to determine a reasonable bend radius.

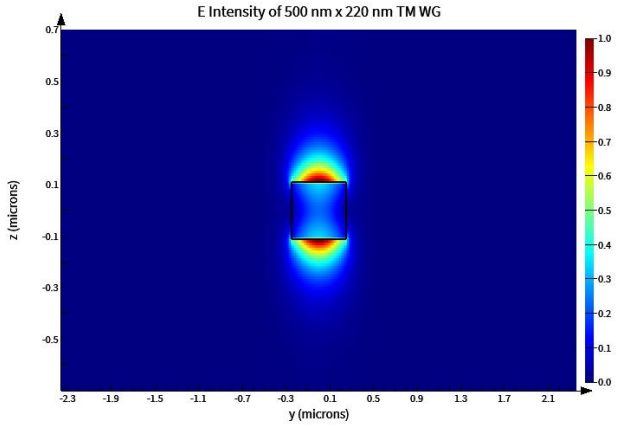


Fig. 5. E intensity of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TM mode 1.

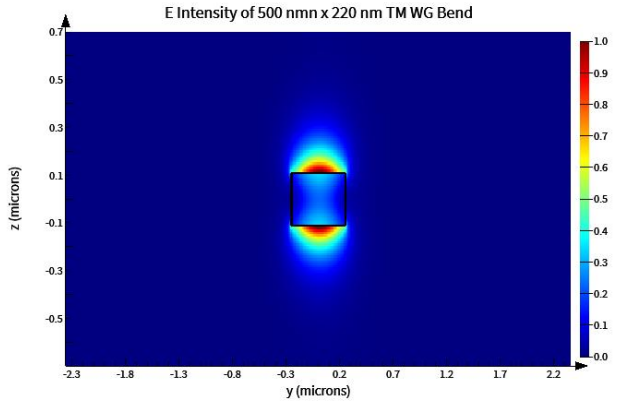


Fig. 6. E intensity of a 500 nm x 220 nm crossection of a bent Si waveguide with bend radius 15 nm in SiO2 cladding of the quasi-TM mode 1.

If a bend radius of 5 nm was used for the quasi-TM mode as it is done in the quasi-TE mode, then there would be no power coupling because the bend is too small to allow the TM mode to propagate. When the bend radius is increased to 10 nm, the power coupling becomes 0.99438, which is 0.6% or XX dB of loss. This can be improved even further with a 15 nm bend radius, which yields a power coupling of 0.997699, which is 0.3% or XX dB of loss. Of course, the bend radius could be even larger to improve the loss even further, but that comes as the cost of floor plan space. Therefore, the MZIs designed for TM mode use a bend radius of 15 nm.

Lumerical MODE was also used to model n_{eff} and n_g of the quasi-TM mode.

IV. Layout

kLayout was used to design the different MZI variants and create the floor plan. The layout consists of four TE MZI variants, two TM MZI variants, two micro-ring resonator variants, and a loopback calibration circuit (to de-embed the GC response).

The three basic TE MZIs composed of two GCs, two Y-branches, and different length waveguides were simulated using Lumerical INTERCONNECT.

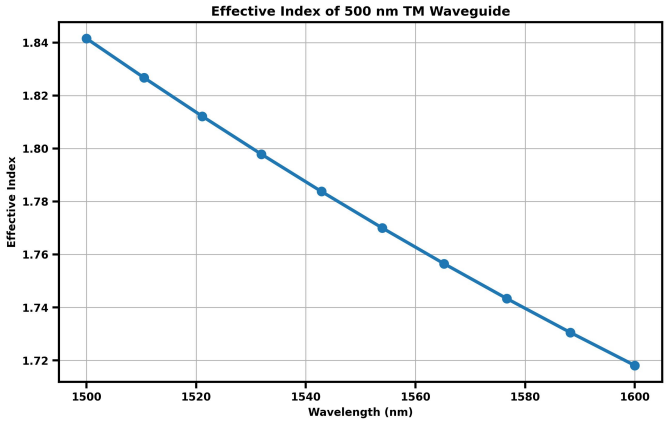


Fig. 7. n_{eff} of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TM mode 1.

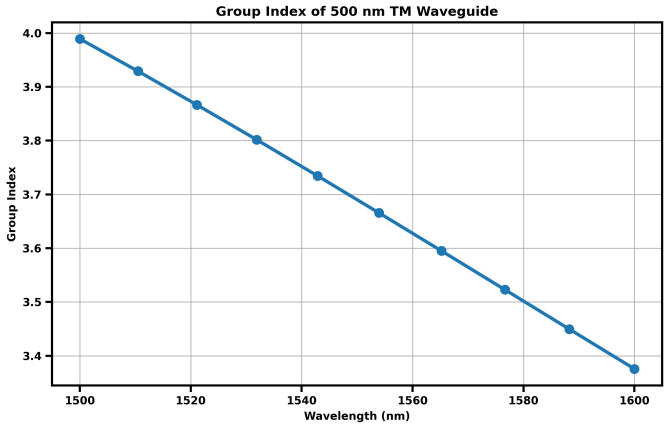


Fig. 8. n_g of a 500 nm x 220 nm crossection of a Si waveguide in SiO2 cladding of the quasi-TM mode 1.

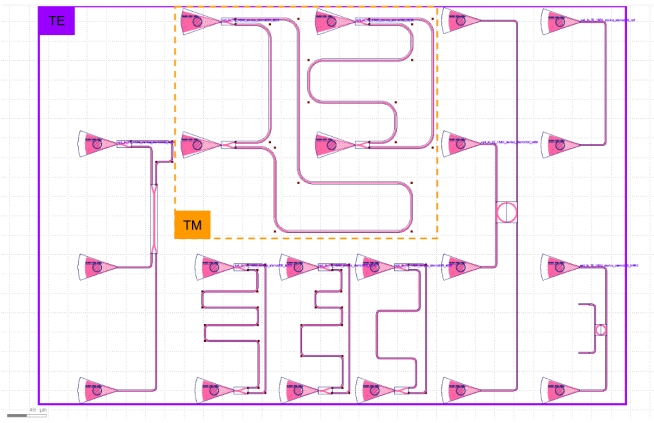


Fig. 9. Floor plan with various MZI devices. Every device in the purple box is TE except for those enclosed in the orange dashed box, which is TM.

A. Quasi-TE Mode MZIs

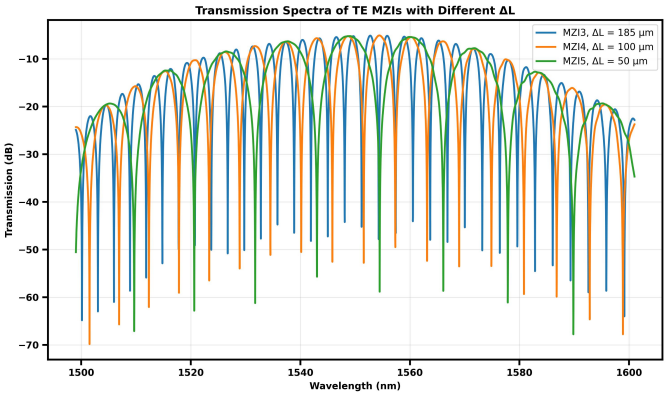


Fig. 10. Spectra of 3 standard TE MZIs with different ΔL.

The FSRs were also simulated in Lumerical. One FSR vs. wavelength plot is shown as an example.

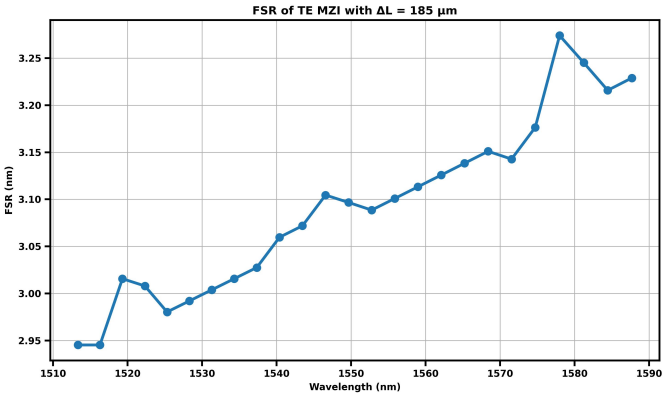


Fig. 11. FSR of MZI3 (TE, $\Delta L = 185um$).

The FSR at the point closest to 1550 nm for each of the normal TE MZI variants was taken and plotted to examine the relationship between FSR and ΔL. As expected, the plot shows an inverse relation where $FSR \propto \frac{1}{\Delta L}$

There is another variant of the MZI which is composed of one input GC connected to a Y-branch and two output GCs connected by a bidirectional coupler.

B. Quasi-TM Mode MZIs

C. FSR

The FSR equation with respect to n_g derived in the Theory section II can be used to find n_g from experimental data. FSR will be measured by observing the transmission spectra of all devices and finding their $\Delta\lambda$ between the two peaks nearest the center wavelength of 1550 nm. Then using the rearranged FSR equation

$$n_g = \frac{\lambda^2}{FSR\Delta L}$$

n_g can be solved, where λ is the wavelength in between the two peaks in question, and ΔL is the difference in length of the imbalanced MZI.

The variants and their expected FSRs and ng are tabulated below.

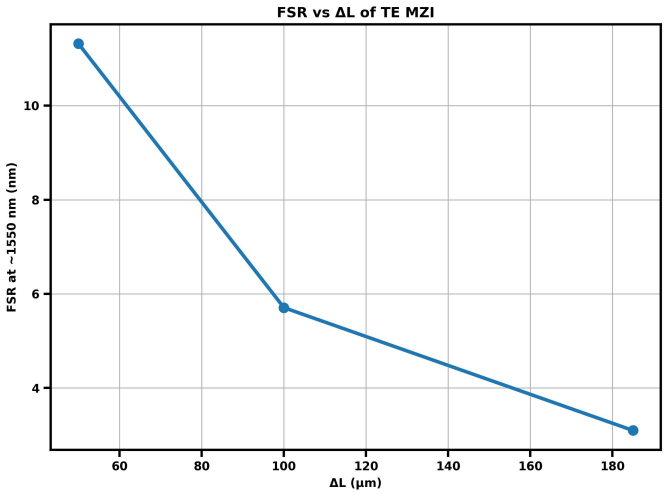


Fig. 12. FSR values vs ΔL

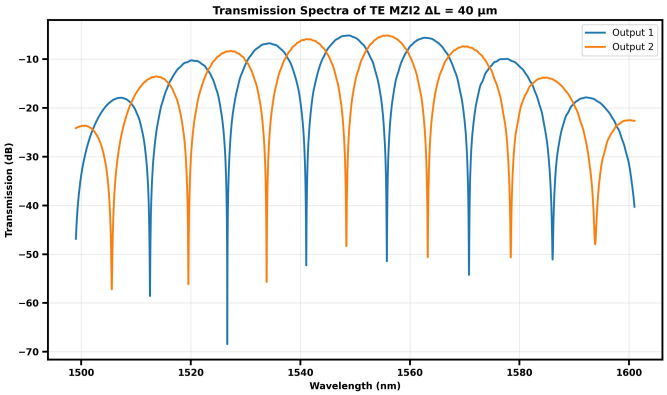


Fig. 13. Spectra of MZI2 (TE, $\Delta L = 40um$.

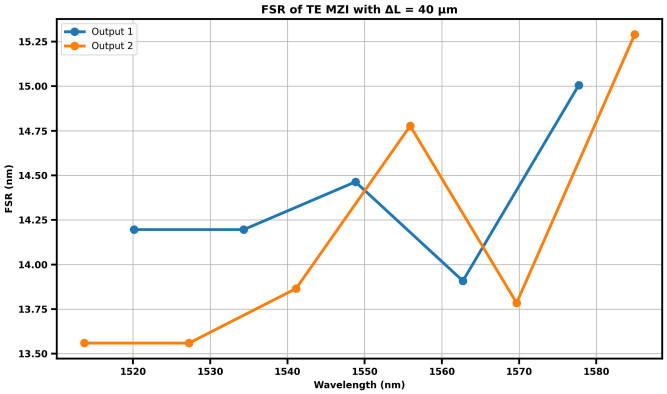


Fig. 14. FSR of TE MZI2

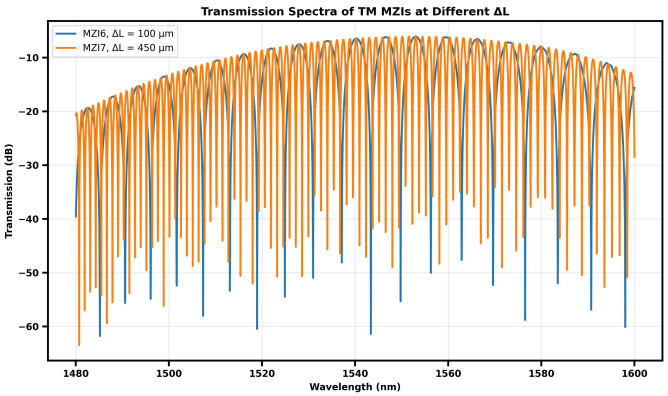


Fig. 15. Spectra of TM MZIs

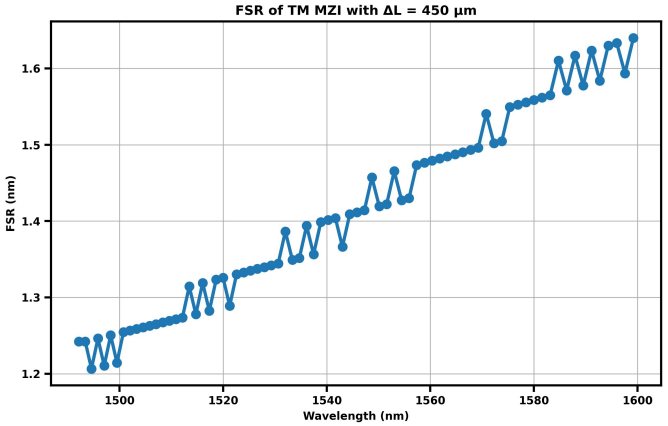


Fig. 16. FSR of MZI7 (TM, $\Delta L = 450um$

V. Fabrication
VI. Experimental Data
VII. Analysis
VIII. Conclusion
Appendix A
Appendix B
Acknowledgment

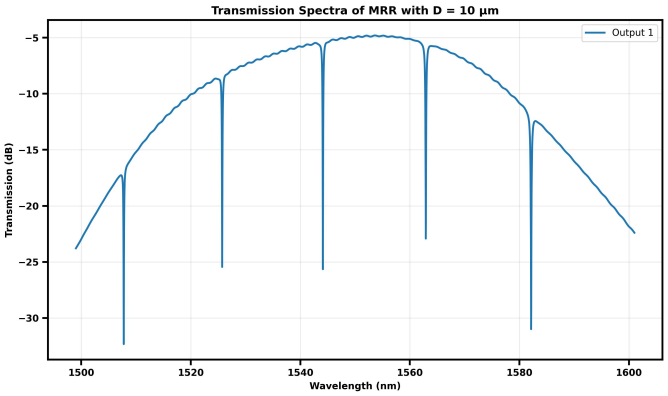


Fig. 17. Spectra of MRR (TE, D = 10 um

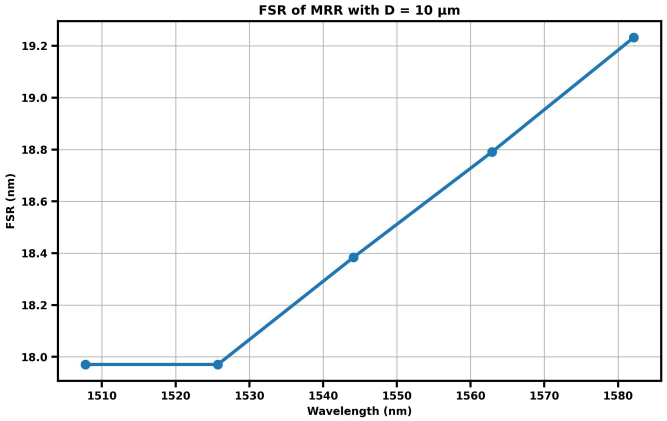


Fig. 18. FSR of MRR (TE, D = 10 um

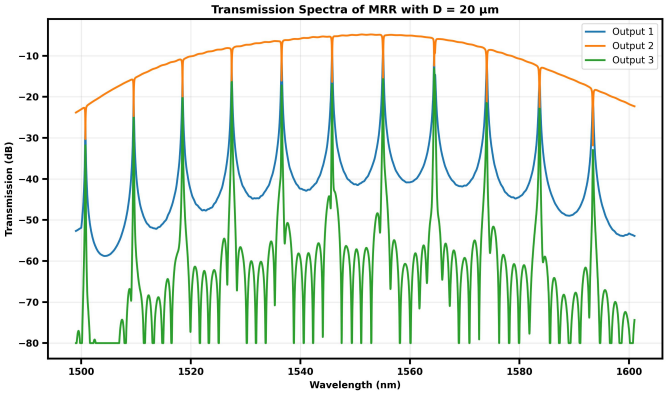


Fig. 19. Spectra of MRR (TE, D = 20 um

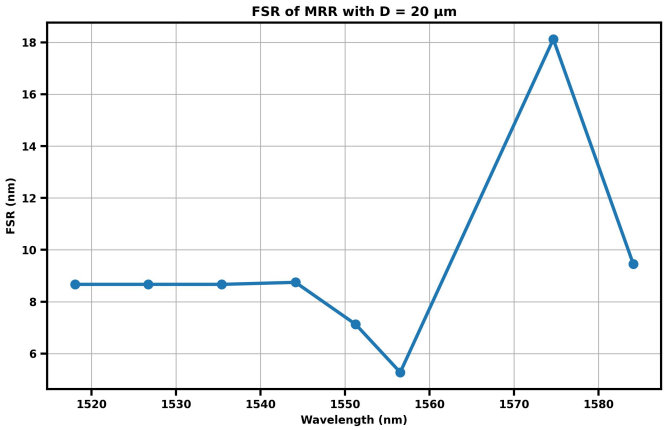


Fig. 20. FSR of MRR (TE, D = 20 um

TABLE I
SUMMARY OF MZI AND MRR VARIANTS AND THEIR METRICS. ASSUMES 1550 NM

Device Name	Variant	Mode	ΔL (μm)	FSR (nm)	n_g
MZI2	BDC	TE	40	14.42	4.17
MZI3	Y-B	TE	185	3.09	4.19
MZI4	Y-B	TE	100	5.71	4.21
MZI5	Y-B	TE	50	11.28	4.26
MZI6	Y-B	TM	100	6.39	3.76
MZI7	Y-B	TM	450	1.42	3.75
MRR	D10	TE	N/A	18.5	4.13
MRR	D20	TE	N/A	8.5	4.50